



Zhongshuo Du

Algorithm Researcher

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Profile

Alumnus of the **State Key Lab of CAD&CG**, Zhejiang University. 4+ years of experience in **AIGC, 3D Vision** and **Geometry Processing**. Proven track record at **MiHoYo** and **Tencent** in developing production-ready pipelines for **3D asset generation** and **automated scene synthesis**. Expert in integrating **MLLMs** into complex industrial workflows. Solid foundation in mathematics and algorithms coupled with high-performance system-building skills, with a focus on bridging cutting-edge research in **Spatial Intelligence & Generative Modeling** with scalable engineering solutions.

Experience

SynWorld (Startup)

Co-Founder / Shanghai / Jun. 2025 – Present

- Co-founded with distinguished professors from HKUST & MUST to bridge advanced CV research with industrial AIGC production.
- Led the development of a controllable video generation platform FizzReel.

MiHoYo | Entropy Lumi Team (T1 Graphics Group)

Graphics Algorithm Researcher / Shanghai / Jun. 2022 – Jun. 2025

- Led R&D for AIGC-based 3D asset generation, automated scene layout, and high-performance graphics algorithms.

Tencent | TiMi Technology Center (Game AI Group)

AI Research Intern / Shenzhen / Jun. 2021 – Aug. 2021

- Researched PCG scene generation algorithms and developed indoor layout generation using Graph Neural Networks and transformers.

Education

Zhejiang University | the State Key Lab of CAD&CG (GAPS)

M.Eng. in Computer Technology / Sep. 2019 – Mar. 2022

- Recommended for admission without Entrance Exam.
- Major in CG&CV, advised by Prof. Youyi Zheng and Prof. Kun Zhou.

Zhejiang University

B.Eng. in Electronic Science and Technology / Sep. 2015 – Jun. 2019

- Student Union Minister & Class Committee Member & Information Officer.
- Achieved 5.0/5.0 GPA in Digital Image Processing, C Programming, Signals and Systems Lab, among others.

Projects

FizzReel: Controllable Video Reimagine Platform

SynWorld(Startup) / Jun. 2025 – Feb. 2026

- Developed a video replication platform focusing on high-precision camera motion replication by integrating **Uni3C** technology on the **Wan2.2** model.
- Architected a modular backend workflow based on **ComfyUI**, supporting automated pipelines from reference video parsing to structured generation.

Multimodal LLM-driven 3D Scene Layout & Synthesis

MiHoYo / Jan. 2025 – Jun. 2025

- Integrated **GPT-4V**, **GroundingDINO**, **SAM**, and **MoGe** for component detection and depth extraction from concept art, generating CSS-formatted layouts after Manhattan World coordinate alignment.
- Built a **Blender** synthesizer via **MCP/Python** API for automated placement, integrating **VLM-based** asset retrieval, **Claude** for spatial reasoning, and an **AIGC-3D** generative fallback, accelerating **Level Artist (LA)** workflows.

AIGC-3D Mesh & Topology Optimization

MiHoYo / Jun. 2024 – Dec. 2024

- Developed a **patch-based segmentation** pipeline using **Watershed** and **Graph-Cut** to smooth boundaries of raw meshes generated from concept art via the **CLAY** model.
- Built a wireframe interaction tool based on **CGAL** half-edge structures to generate production-ready **quad-dominant** meshes.

3D Stylized Stone Generation System

MiHoYo / Oct. 2022 – Jun. 2024

- Developed a multi-view stylized stone images generation pipeline using **LoRA-tuned Stable Diffusion** controlled by **ControlNet/T2i-Adapter**, implemented an **Inpainting-based strategy** to ensure strict multi-view consistency for stylized stone images.
- Leveraged **GANs** for precise **normal map** estimation from 2D images, reconstructed high-poly stylized stone assets via **density-adaptive large-steps differential rendering** combined with **continuous-remeshing**.



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Skills

- Prog:** C++ & Python & JavaScript
- DL:** Pytorch & Tensorflow
- CG:** OpenGL & OpenCV & CGAL & Open3D & Libigl & Eigen
- DCC:** Blender & Houdini & Zbrush & Maya & 3dsMax & SP & SD
- Engine:** UE & Unity
- Lang:** English & Mandarin
- Misc:** Cursor & Trae & OpenClaw

Interests



Swimming



Boxing



Music



Game



Calligraphy



Photography

Projects continued

LOD Visibility Weight and Culling

MiHoYo / Jun. 2022 – Oct. 2022

- Enhanced the **QEM-based LOD** algorithm by adding visibility weights and culling logic to intelligently control polygon reduction priority for large-scale scenes.

PCG Scene Generation

Tencent / Jun. 2021 – Aug. 2021

- Implemented **graph**-based layout generation and utilized **transformers** for automated indoor scene object placement and orientation.

Teeth Alignment Prediction in Videos

ZJU / Sep. 2021 – Mar. 2022

- Utilized **U-Net** for tooth segmentation followed by frame-by-frame **pose fitting** of 3D orthodontic models to extract target geometric priors.
- Developed a **spatio-temporal VAE-GAN** with **FlowNet** motion warping to synthesize high-fidelity and temporally stable orthodontic preview videos.

Single-view 3D Hair Generation

ZJU / Dec. 2020 – Jun. 2021

- Designed a **Voxel-UNet** architecture to predict 3D **occupancy** and **orientation** fields from monocular images.
- Generated **polystrip** hair models by iteratively growing strands along the predicted orientation field using a greedy expansion strategy.

Mesh Reconstruction from Noise Point Cloud

ZJU / Mar. 2019 – Nov. 2020

- Extracted geometric features via **Normal Tensor Voting** and converted them into graph representations.
- Trained a **GCN** to cascade-predict guided normals for iterative vertex updates. Research published in **ACM Transactions on Graphics (TOG)**.

Hand Tracking and 3D Scene Interaction

ZJU / Jul. 2018 – Oct. 2018

- Regressed low-dimensional hand joint poses from downsampled 3D point clouds using **PointNet** with fingertip refinement.
- Developed a real-time interaction system in **Unity** for **Leap Motion** devices to enable natural hand-to-virtual-scene interaction.

Selected Publications

- Zekai Gu, Shuoxuan Feng, Yansong Wang, Hanzhuo Huang, **Zhongshuo Du**, Chengfeng Zhao, Chengwei Ren, Peng Wang, Yuan Liu, "GO-Renderer: Generative Object Rendering with 3D Asset-aware Controllable Video Diffusion Models", ECCV 2026 (submission).
- Hanzhuo Huang, Qingyang Bao, Zekai Gu, **Zhongshuo Du**, Cheng Lin, Yuan Liu, Sibe Yang, "RefAny3D: 3D Asset-Referenced Diffusion Models for Image Generation", ICLR 2026.
- Yuefan Shen, Hongbo Fu, **Zhongshuo Du**, Xiang Chen, Evgeny Burnaev, Denis Zorin, Kun Zhou, Youyi Zheng, "GCN-Denoiser: Mesh Denoising with Graph Convolutional Networks", ACM Trans. Graph., 2022.

Honors & Awards

- Excellent Postgraduate Students 2022
- Triple-A Graduate 2020-2021
- Award of Honor 2020-2021
- Outstanding Graduation Thesis 2019
- Excellent League Member 2017-2018
- Outstanding Students Scholarship 2016-2018
- Outstanding Merits Scholarship 2016-2018
- National Talents Training Base Scholarship 2016-2018
- Mathematics Olympiad Second Prize 2014
- NOI Member of the Provincial Team 2014
- NOIP First Prize 2013